

Programme	B. Sc. Computer Science				
Course Title	Storage Technologies				
Type of Course	Elective				
Semester	VI				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	9. Basic knowledge of computer systems and architecture 10. Fundamental understanding of data structures and algorithms				
Course Summary	This course introduces students to various storage technologies, storage network technologies, storage and virtualization technologies. Course also discuss various back up and recovery strategies.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand fundamentals of Information storage	U	C	Instructor-created exams / Quiz
CO2	Examine features of various storage architectures	U	C	Assignment / Seminar presentations/ Exams
CO3	Understand features of Intelligent storage systems	U	P	Seminar Presentation / Group Tutorial Work/ Viva Voce
CO4	Identify features of various Storage technologies	U	C	Instructor-created exams / Home Assignments
CO5	Identify need of backup and recovery and various recovery mechanisms	U	P	Writing assignments/ Exams/ Seminar Presentations
CO6	Infer security needs and management needs for storage technologies	U	F	Case Study/ Exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48+12)	Marks (70)
I	Storage System		12	18
	1	Introduction to Information Storage- Information Storage, Evolution of Storage Architecture	2	
	2	Data Center Infrastructure and characteristics	1	
	3	Third platform technologies- Cloud storage and its characteristics	2	
	4	Cloud services and deployment models	3	
	5	Storage Architectures- Direct-Attached Storage (DAS) Network-Attached Storage (NAS) (Introduction only)	2	
	6	Storage Area Network (SAN) Cloud storage architectures(Introduction only)	2	
II	Intelligent Storage Systems & RAID		12	18
	7	RAID Implementation Methods, RAID Array Components, RAID Techniques	2	
	8	RAID Levels, RAID Impact on Disk Performance	3	
	9	RAID Comparison	1	
	10	Components of an Intelligent Storage System	1	
	11	Storage Provisioning	2	
	12	Types of Intelligent Storage Systems	3	
III	Storage Networking Technologies - Fibre Channel Storage Area Networks		12	18
	13	Block based stored system, File based storage system, object oriented based storage system (Introduction)	2	
	14	Fibre Channel Storage Area Networks- Components of FC SAN,	2	
	15	Fibre Channel Architecture	2	
	16	Fabric Services	2	
	17	FC SAN Topologies	2	
	18	Virtualization in SAN	2	
IV	Backup and Archive		12	16
	19	Backup Purpose, Backup Considerations, Back up Granularity	3	
	20	Recovery Considerations , Backup Methods	3	
	21	Backup Architecture, Backup Topologies	3	
	22	Backup and Restore Operations	3	
V	Open Ended Module		12	
	1	Storage Security Domains	3	
	2	Security Implementations in Storage Networking	3	
	3	Securing Storage Infrastructure in Virtualized and Cloud Environments	3	
	4	Storage Infrastructure Management Activities	3	

References

- Information Storage and Management: Storing, Managing, and Protecting Digital Information in Classic, Virtualized, and Cloud Environments, 2nd Edition, Willey Publications

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	-	2	--	-	1	1						
CO 2	-	2	-	-	1	1						
CO 3	-	1	-	-	1	1						
CO 4	-	1	-	-	2	1						
CO 5	-	1	-	-	2	1						
CO 6	-	-	-	-	2	2						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	End Semester Examinations
CO 1	✓		✓
CO 2	✓		✓
CO 3	✓		✓
CO 4	✓	✓	✓
CO 5	✓	✓	✓
CO 6	✓	✓	